8. (A) Predict the conditions under which

   (i) Aluminium might be expected to reduce magnesia.

   (ii) Magnesium could reduce alumina.

   (B) Carbon monoxide is more effective reducing agent than carbon below 983K but, above this temperature, the reverse is true –Explain.

   (c) it is possible to reduce Fe_2O_3 by coke at a temperature around 1200K

9. Give the uses of zinc.

10. Explain the electrometallurgy of aluminium.

11. Explain the following terms with suitable examples.

    (i) Gangue (ii) slag

12. Give the basic requirement for vapour phase refining.

13. Describe the role of the following in the process mentioned.

    (i) Silica in the extraction of copper.

    (ii) Cryolite in the extraction of aluminium.

    (iii) Iodine in the refining of Zirconium.

    (iv) Sodium cyanide in froth floatation.

14. Explain the principle of electrolytic refining with an example.

15. The selection of reducing agent depends on the thermodynamic factor: Explain with an example.

16. Give the limitations of Ellingham diagram.

17. Write a short note on electrochemical principles of metallurgy.

XII U1 Metallurgy - Jerald.indd 24 3/29/2019 6:34:52 PM